

Statistical learning theory

Foundations of AI

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

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Section 1

Empirical risk minimisation

Vladimir Vapnik



Vladimir Vapnik



Second edition, 2006

Estimation of Dependences Based on Empirical Data. Russian original 1979, Kotz translation Springer 1982, second edition 2006.

Vladimir Vapnik



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- **Catching spam.** The score counts how often the answer is wrong.
- **Fitting a price.** The score is the squared miss, the one this room has used all day.
- **Estimating a probability.** The score is how surprised the model is by what actually happened.
- Three problems that had lived in different books. The score says which one you are doing, and what happens to the score afterwards never changes.

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2

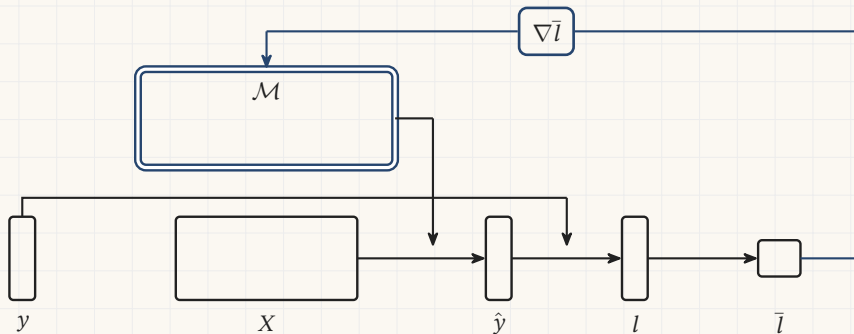
Section 2

The same loop, elsewhere

The loop, with the model as one box

The same picture, with $\mathcal{M}$ where the network was.

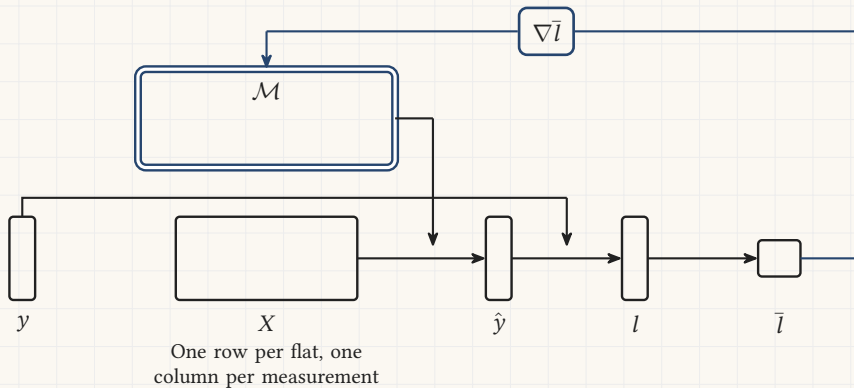
The five flats are the ones from this morning, and the network is the one from this afternoon.



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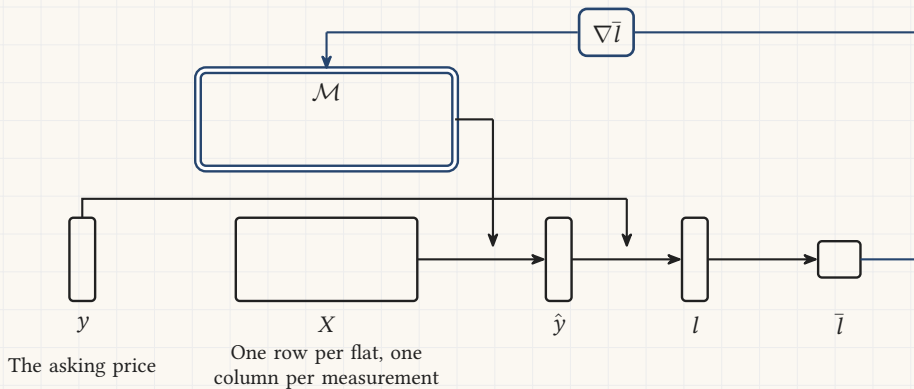
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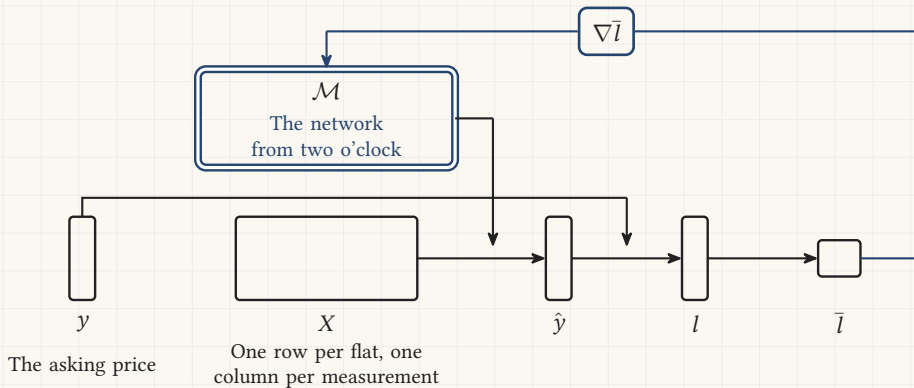
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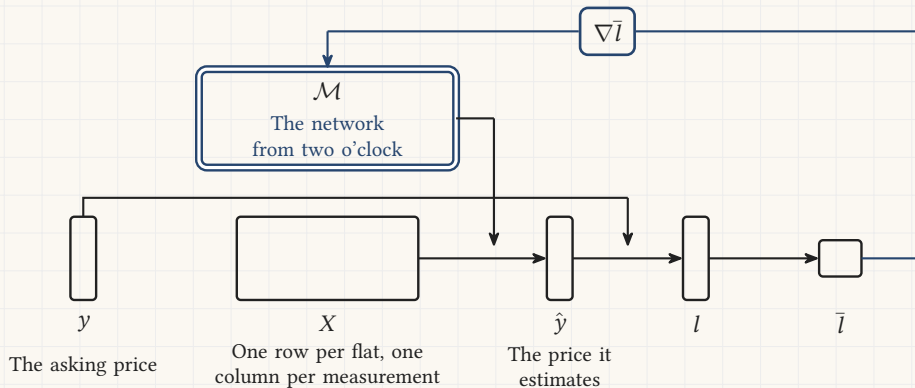
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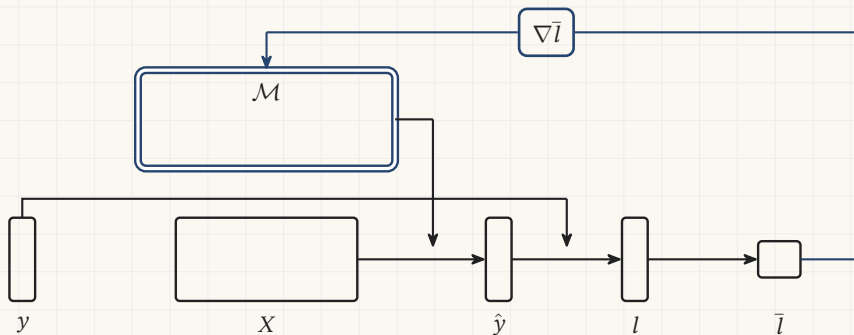
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Large language models

A row is text with its next token cut off. Every position of every document gives one.

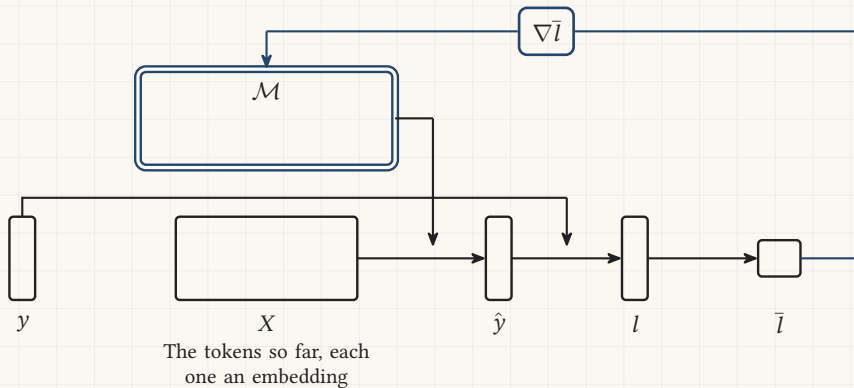
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



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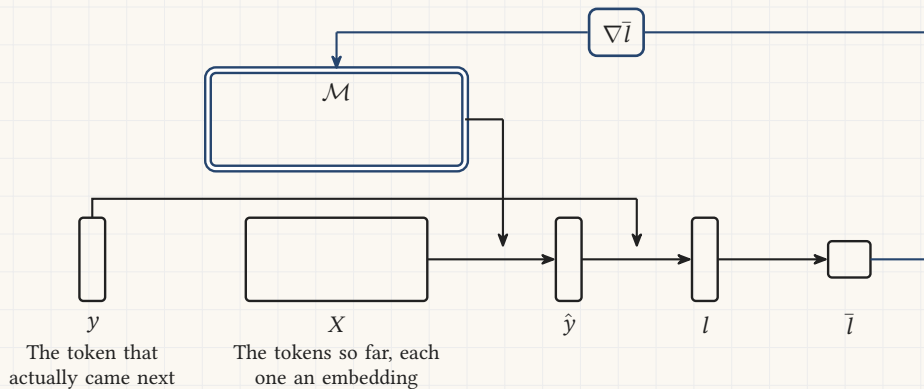
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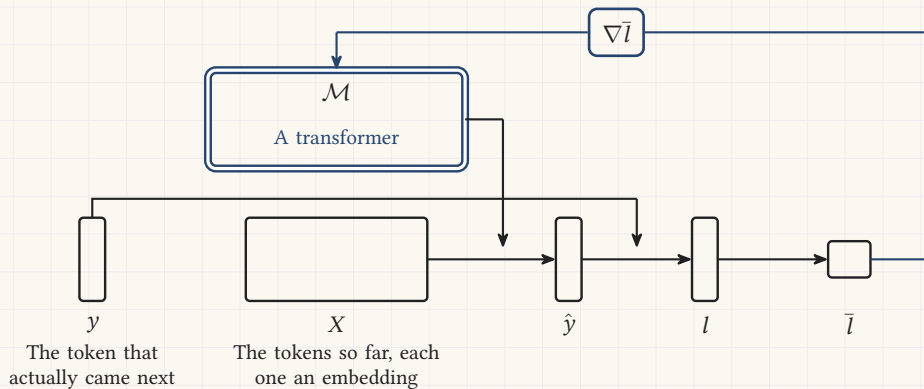
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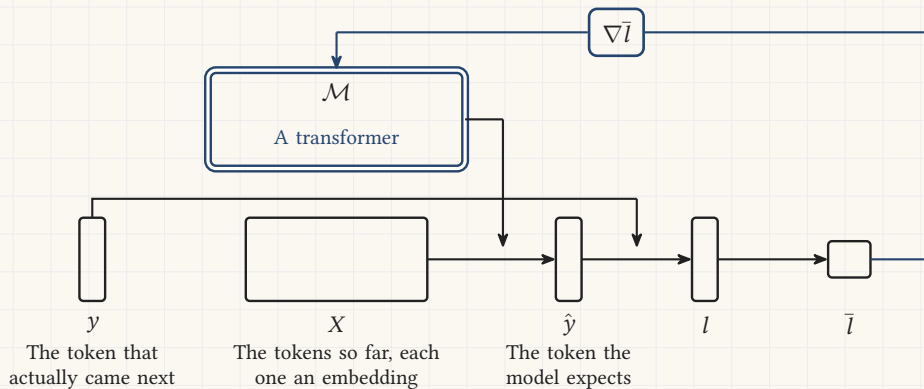


Image generation

Add a measured amount of noise to a picture, then ask which noise was added.

The loss here really is the squared miss. Ho, Jain and Abbeel, 2020; Rombach and others, 2022.

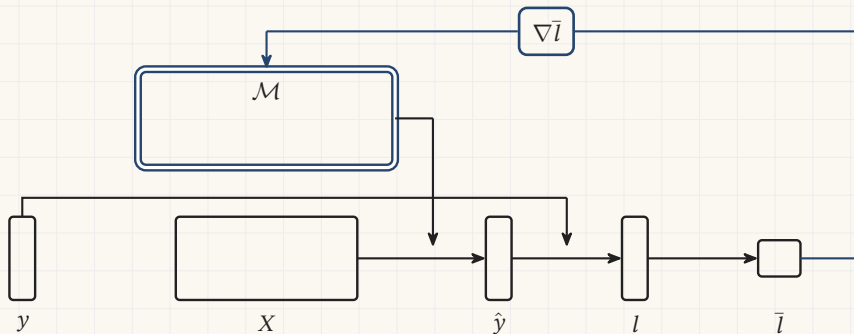


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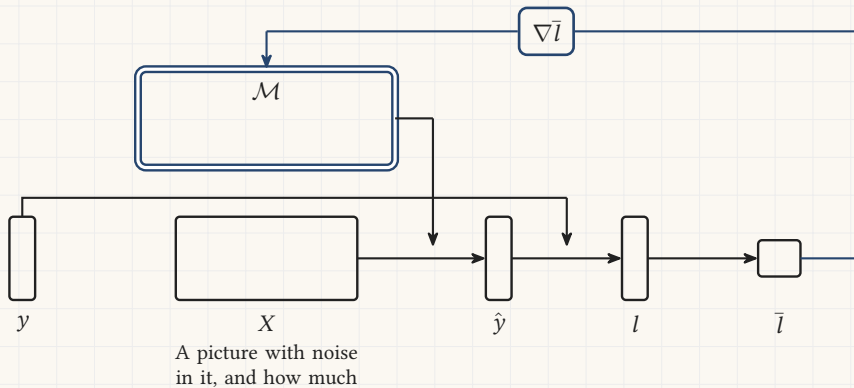


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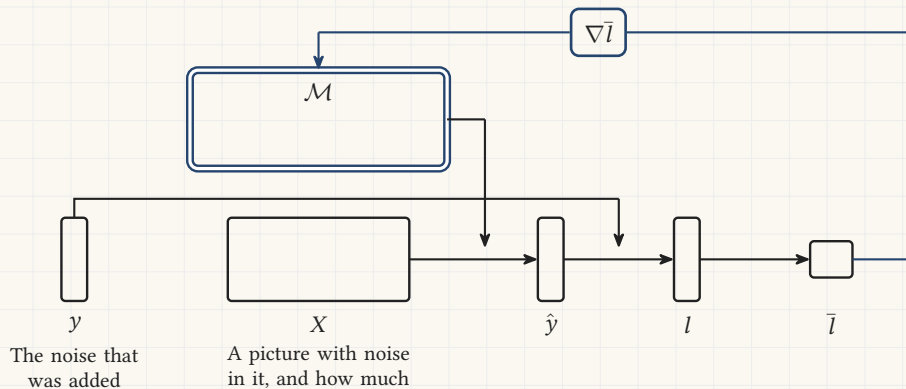


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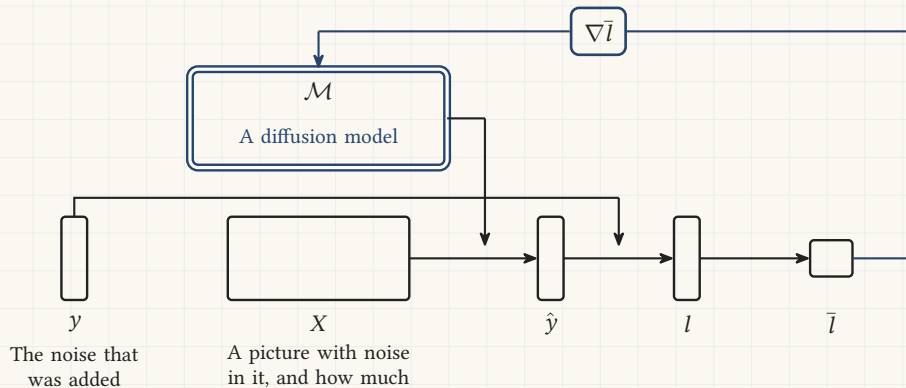
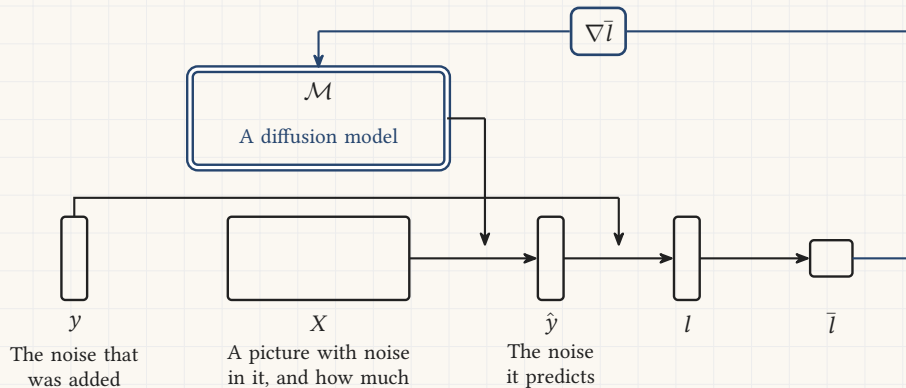


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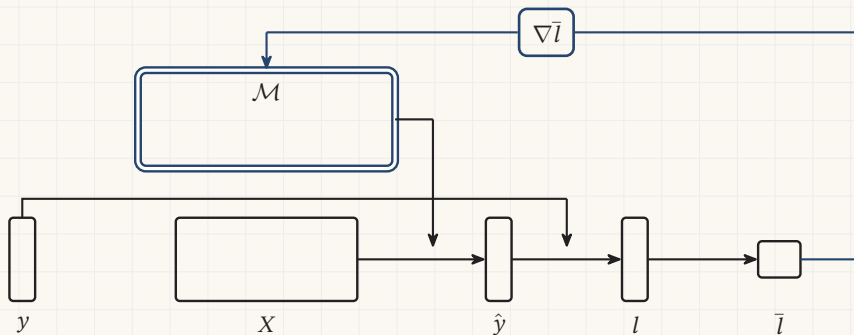
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Transcription

Every recording that already carries a transcript is a row. There are 680 000 hours.

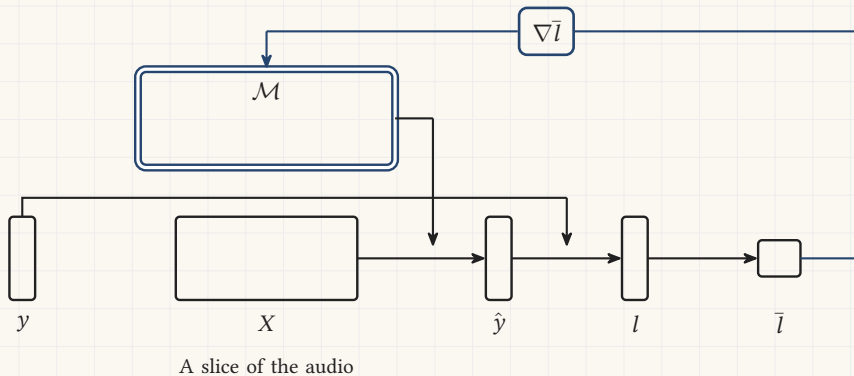
Whisper. Radford and others, ICML 2023.



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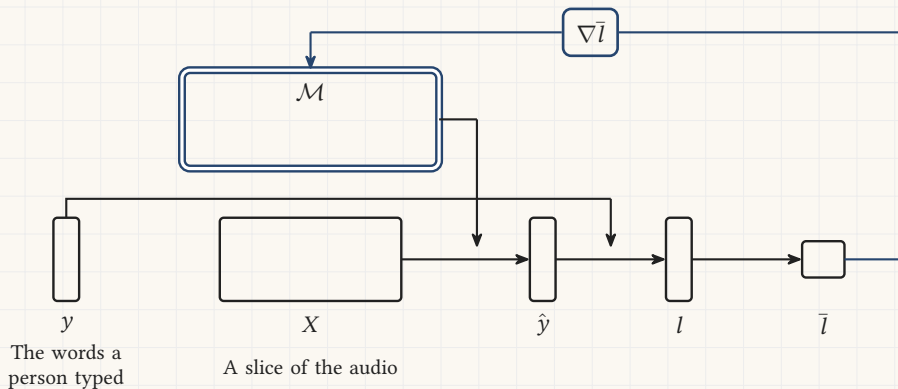
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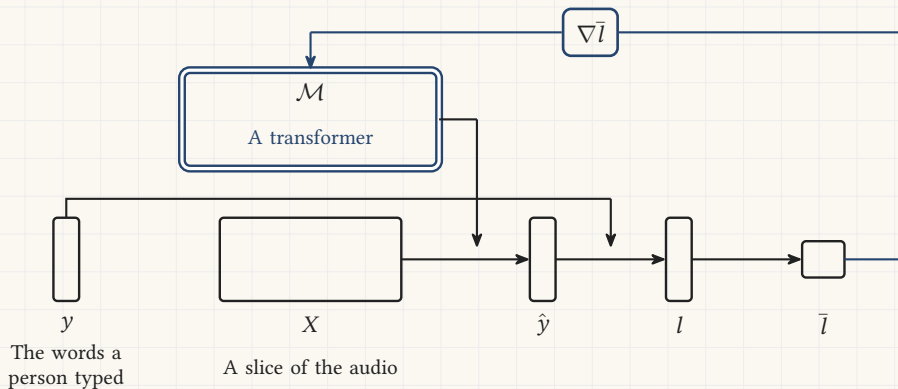
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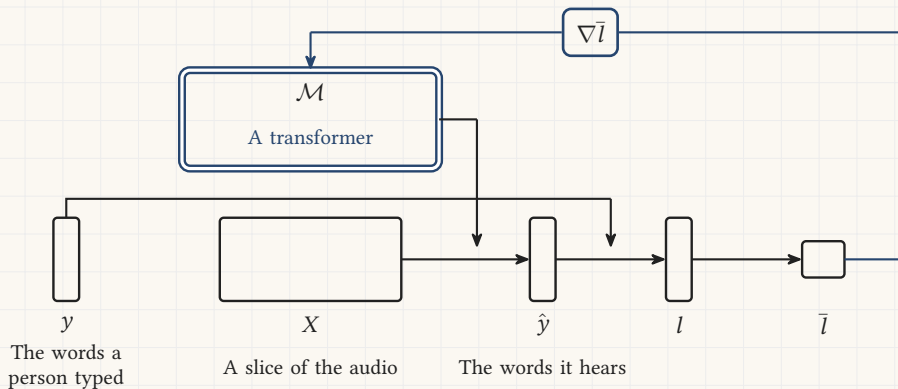
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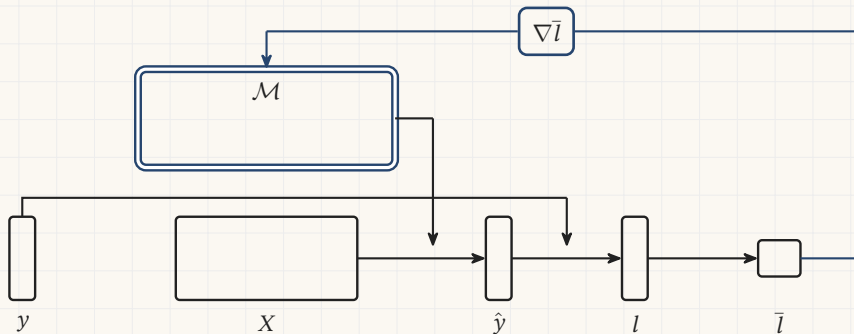
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Ranking and recommendation

Every impression is a row: what was shown, to whom, and whether they took it.

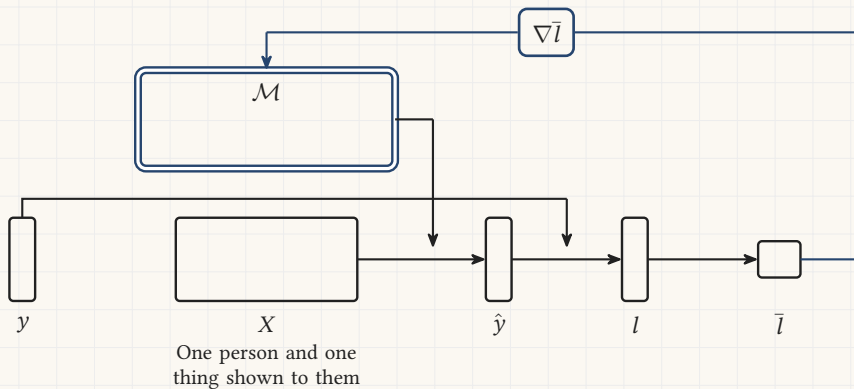
The system that picks the next video. Covington, Adams and Sargin, RecSys 2016.



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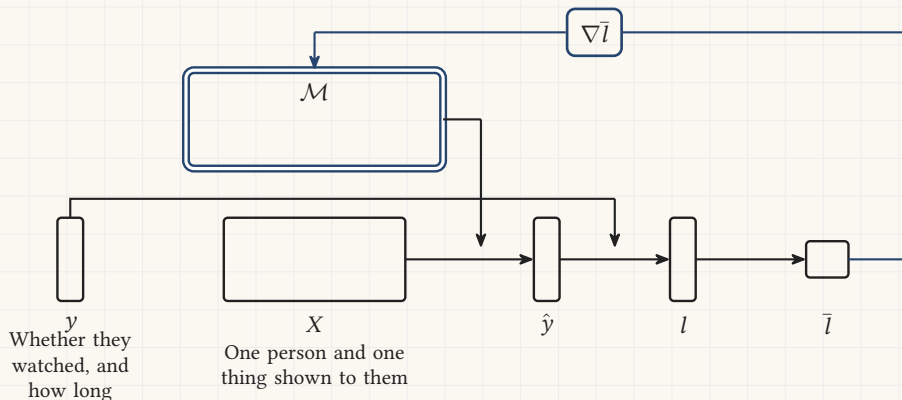
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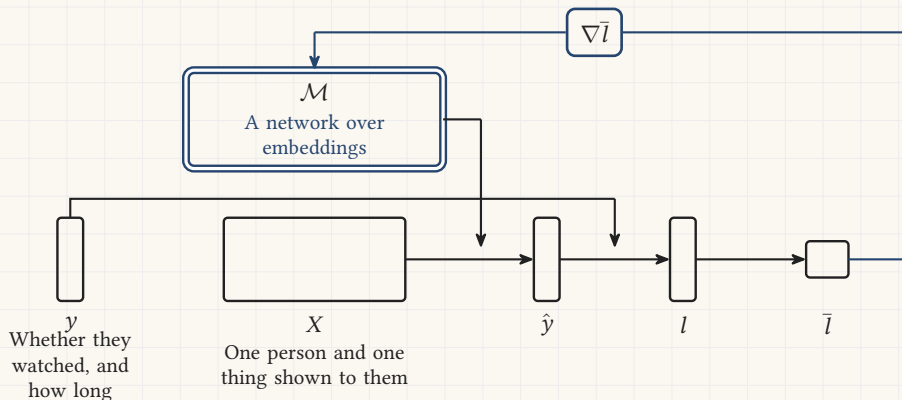
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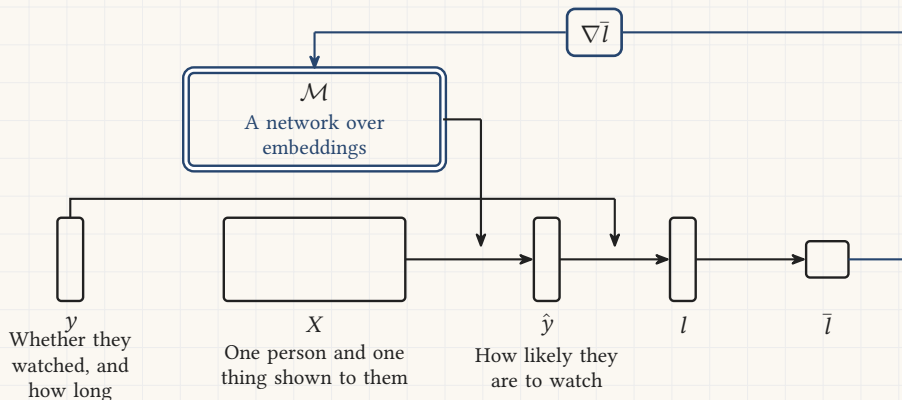
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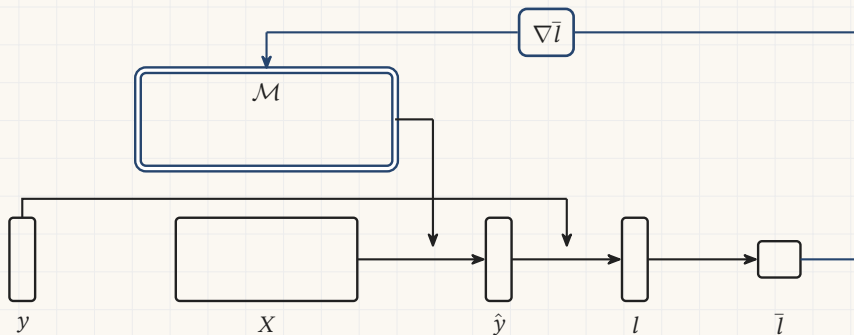
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Fraud and credit decisions

Every settled transaction is a row, and its label arrives weeks later with the chargeback.

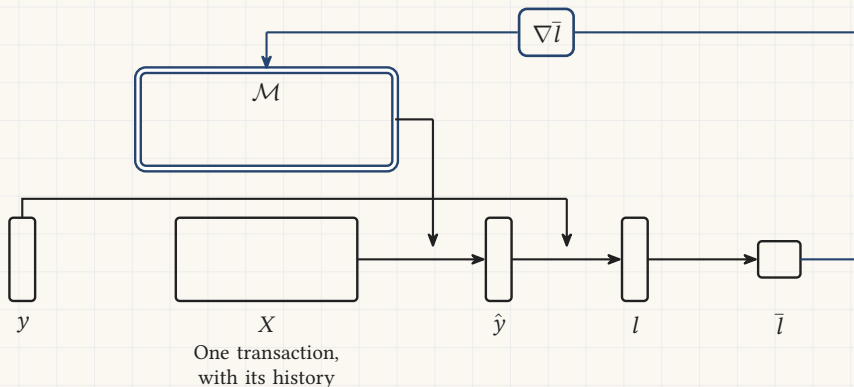
Dal Pozzolo and others, 2018, with Worldline. The tree ensemble is fitted to the gradient too.



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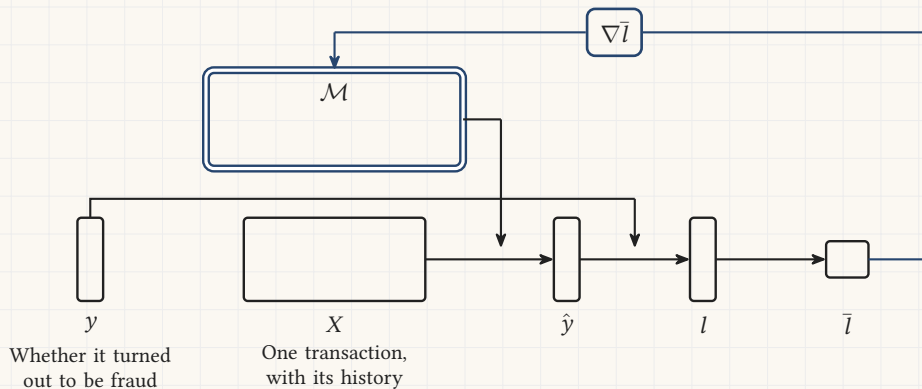
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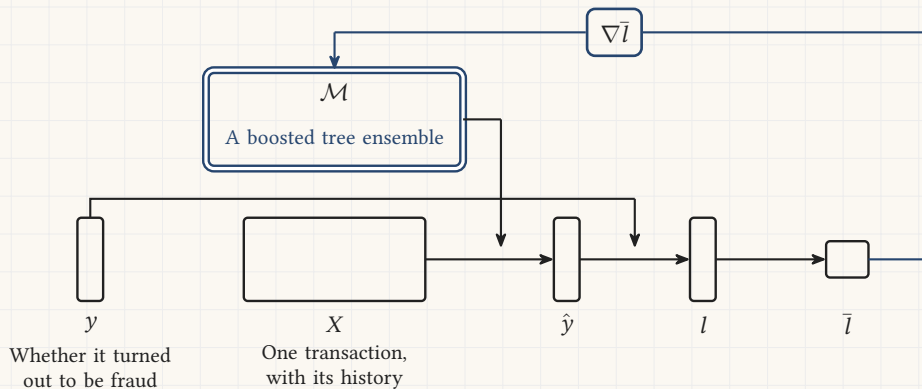
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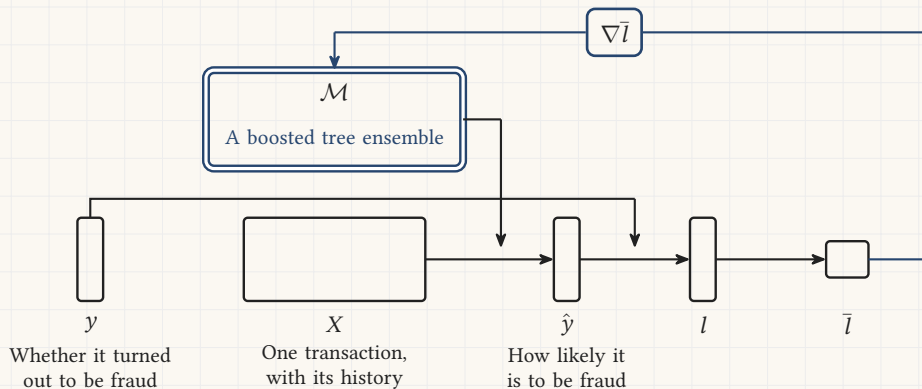
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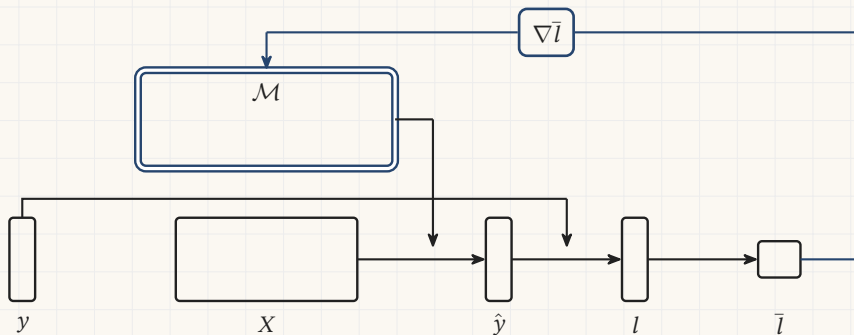
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Screening for diabetic eye disease

In 2018 a regulator cleared a machine to read these photographs with no doctor behind it.

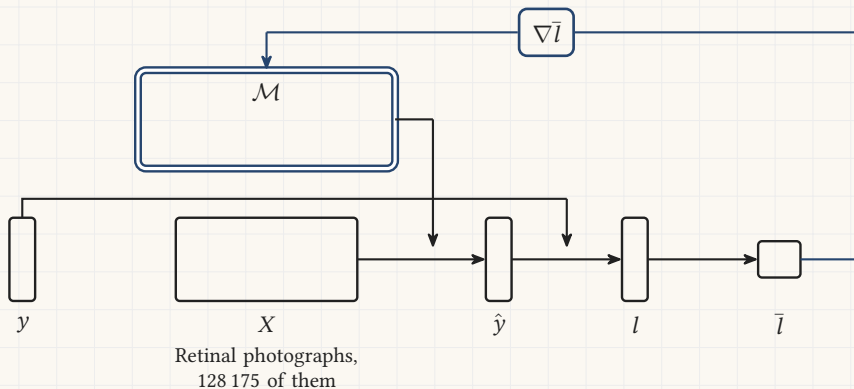
128 175 photographs, graded 3 to 7 times by 54 ophthalmologists. Gulshan and others, *JAMA*, 2016.



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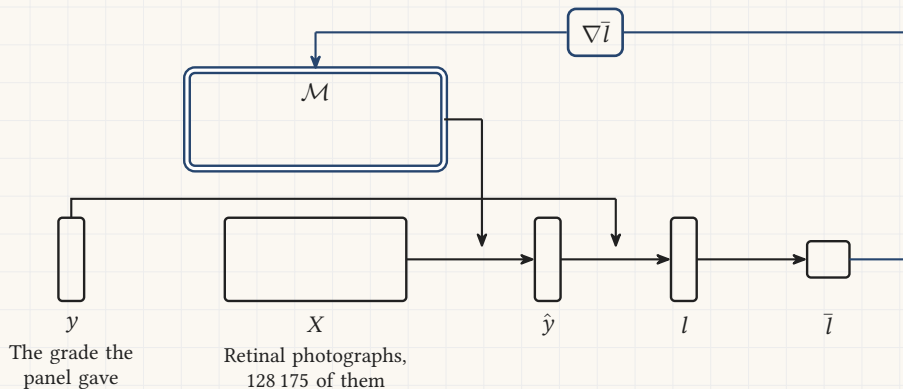
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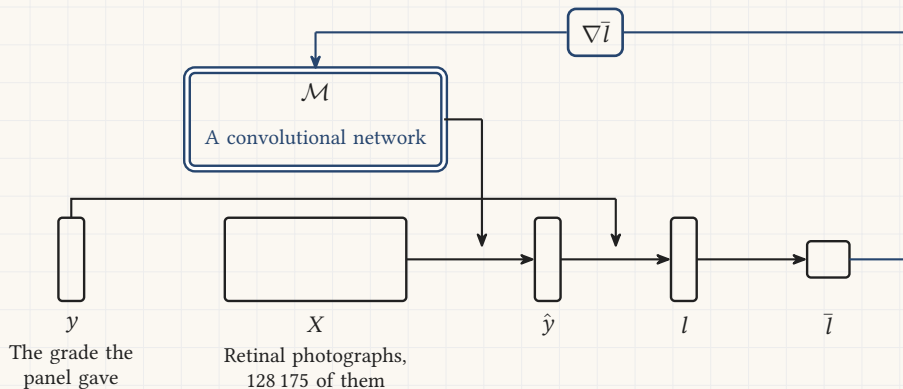
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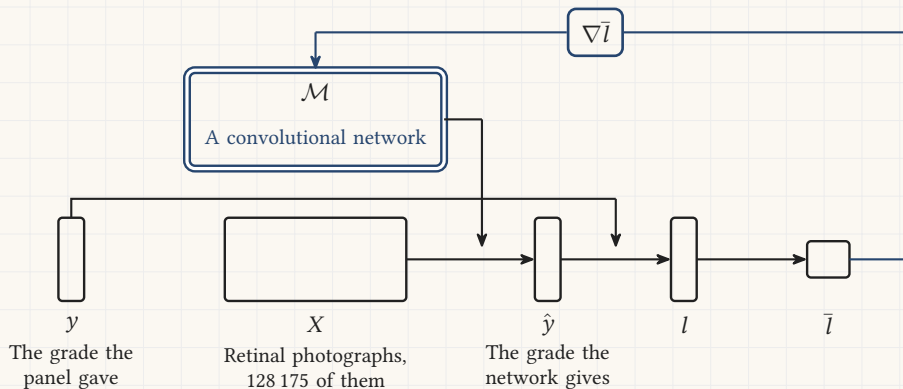
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From LLMs to reasoning models

Agentic AI, part 1

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UZH CAS Gamechanger AI

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Section 1

How a language model is trained

One row for every position

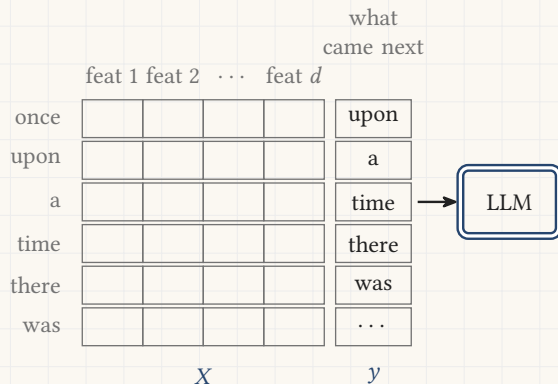
	feat 1	feat 2	...	feat d
once				
upon				
a				
time				
there				
was				

X

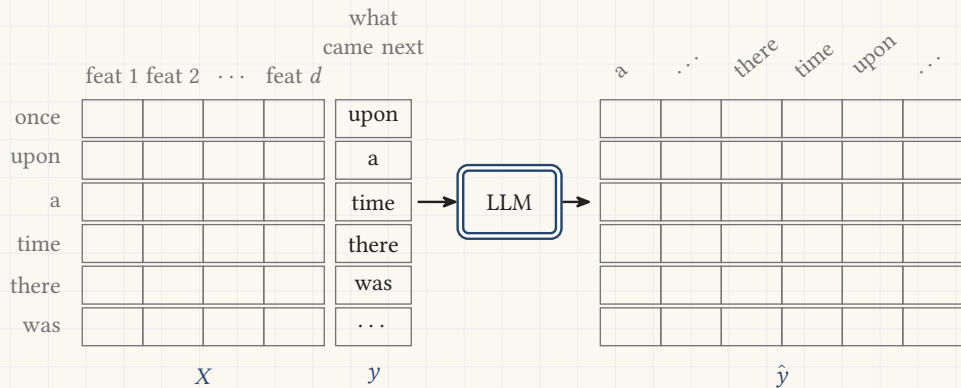
One row for every position

	what came next			
	feat 1	feat 2	...	feat d
once				upon
upon				a
a				time
time				there
there				was
was				...
	X			
	y			

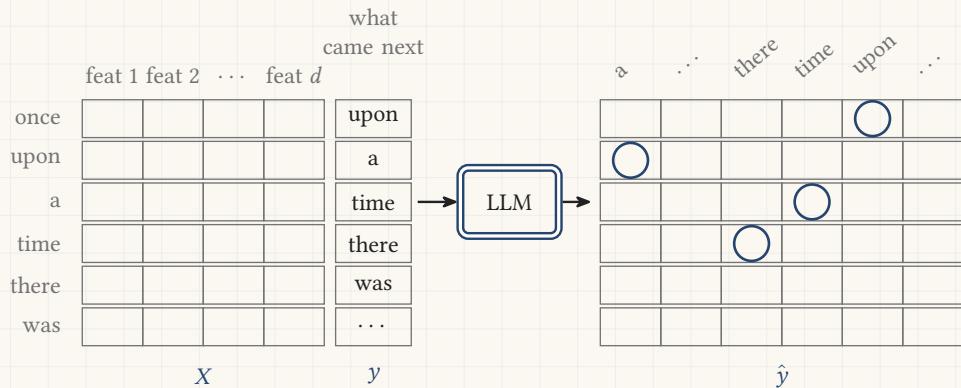
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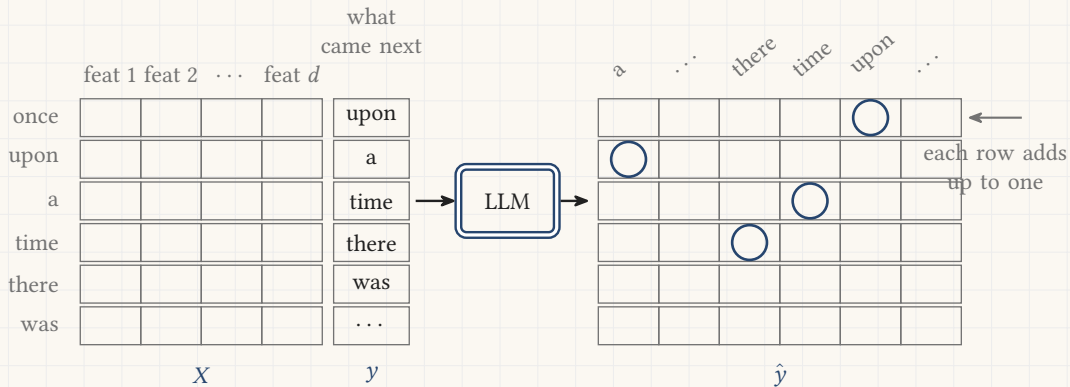
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2

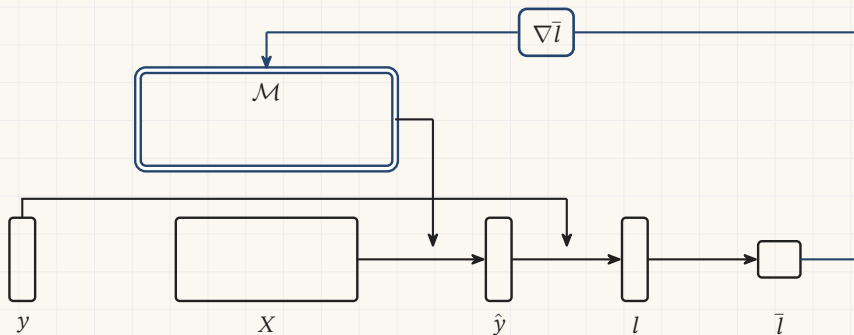
Section 2

How a language model becomes a chatbot

Instruction tuning

GPT-3 was finished in 2020 and it completed text. A smaller second pass made it answer.

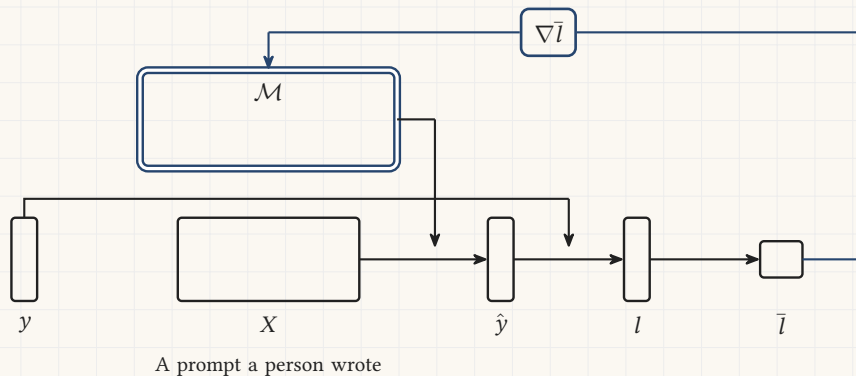
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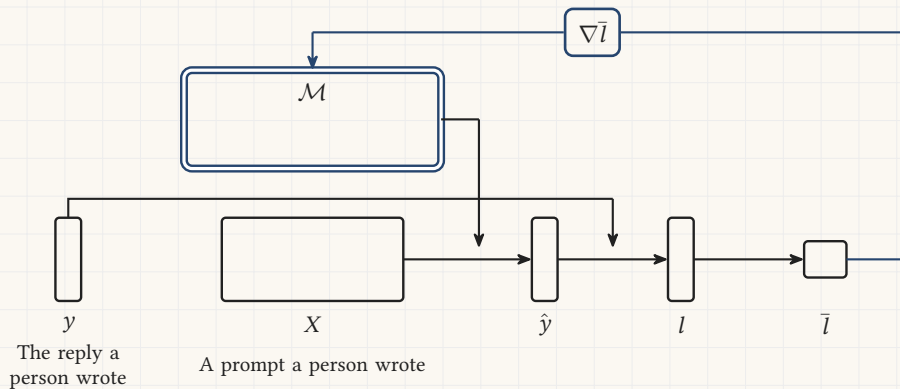
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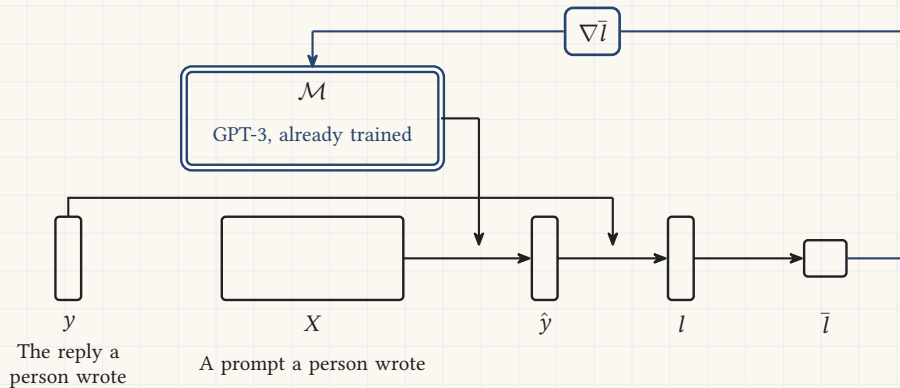
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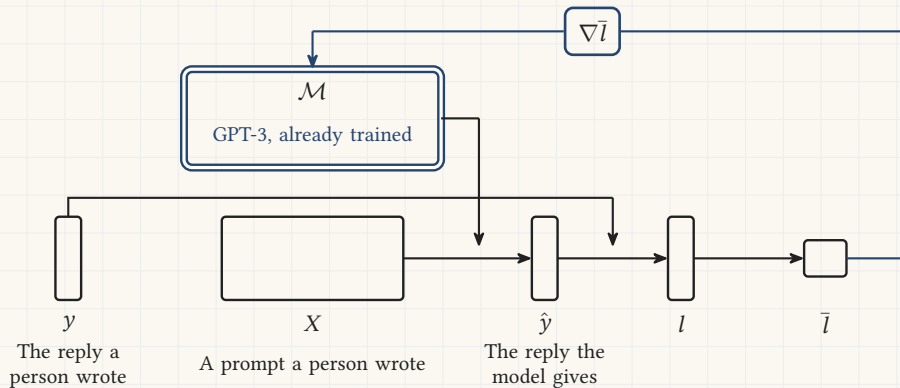
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The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

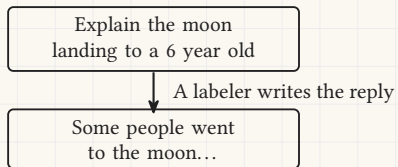
Explain the moon
landing to a 6 year old

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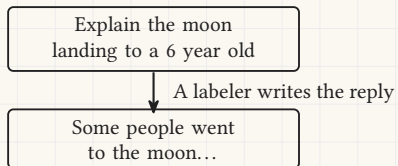
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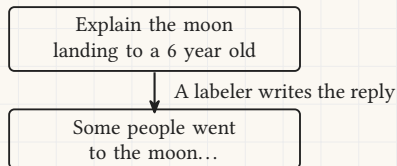
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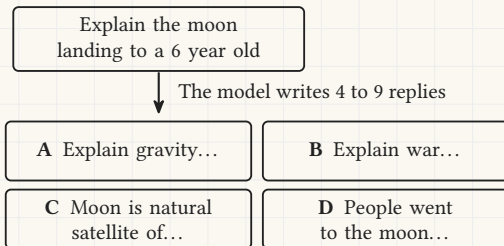
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One comparison



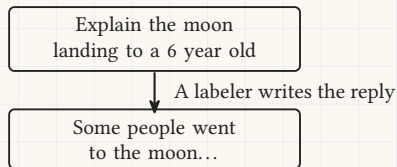
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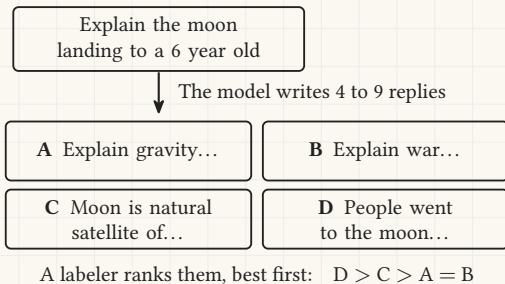
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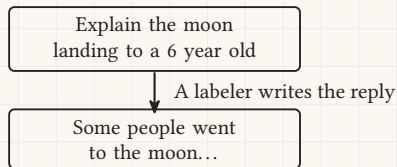
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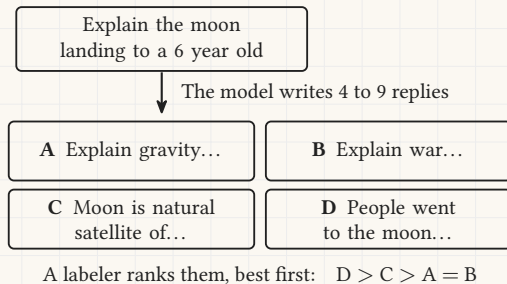
One demonstration

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One comparison

33 207 prompts, most of them sent by users of the API



3

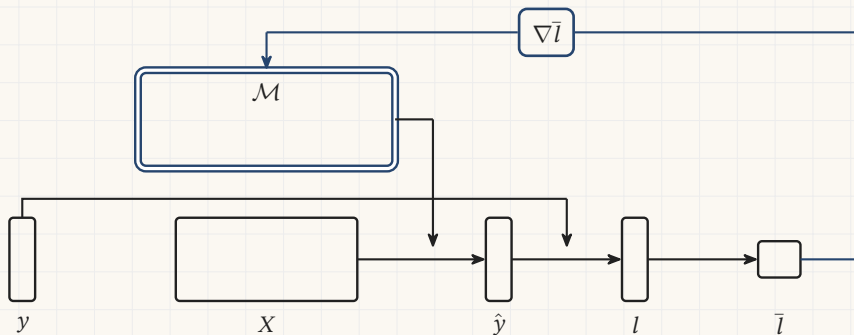
Section 3

How a chatbot becomes a reasoning model

Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

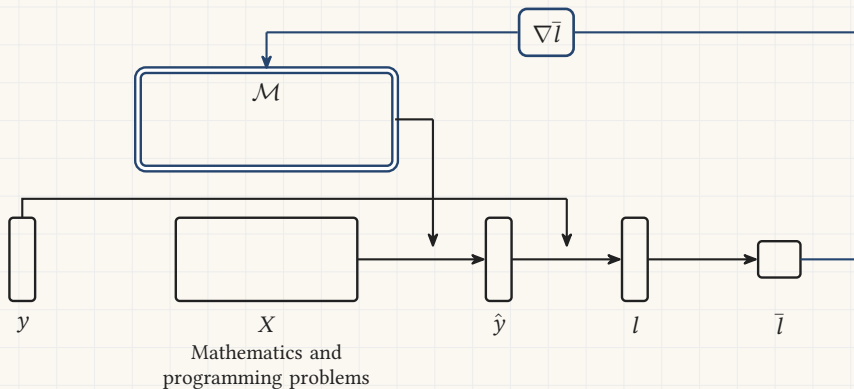
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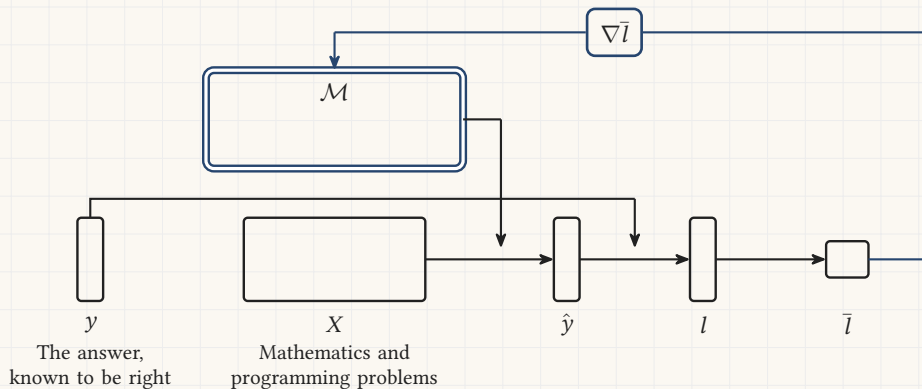
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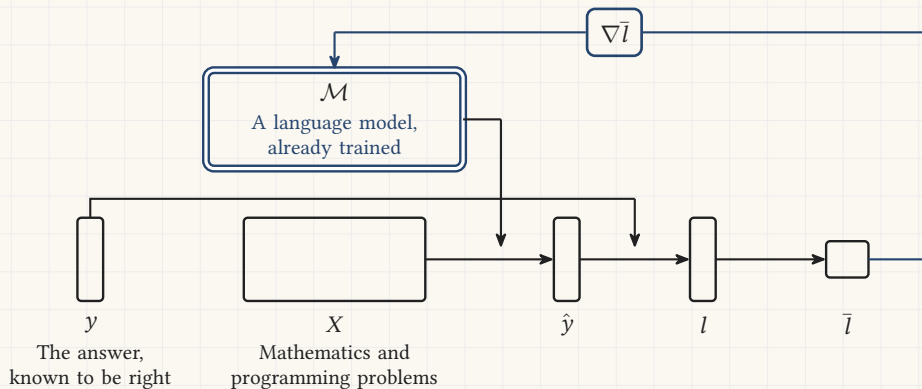
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Reasoning models

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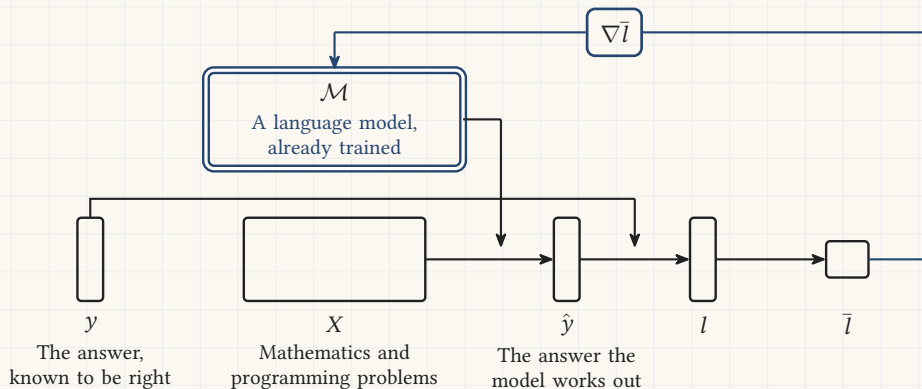
The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



Reasoning models

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4

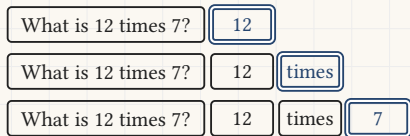
Section 4

Reinforcement learning

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

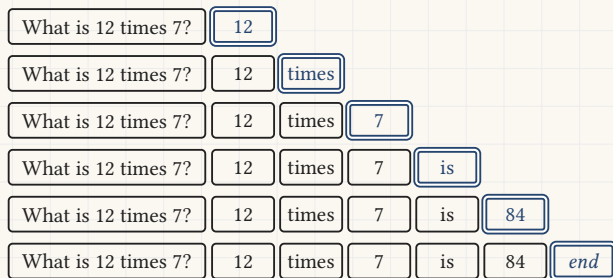
DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

What is 12 times 7?	12				
What is 12 times 7?	12	times			
What is 12 times 7?	12	times	7		
What is 12 times 7?	12	times	7	is	
What is 12 times 7?	12	times	7	is	84

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).



The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	end

0 while the answer is still being written

The end-of-answer token: the episode ends

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

							Reward	
What is 12 times 7?	12						0	
What is 12 times 7?	12	times					0	
What is 12 times 7?	12	times	7				0	0 while the answer is still being written
What is 12 times 7?	12	times	7	is			0	
What is 12 times 7?	12	times	7	is	84		0	
What is 12 times 7?	12	times	7	is	84	end	1	1 at the end: 84 is right. A wrong answer earns 0
The end-of-answer token: the episode ends								

The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

DeepSeek-R1 was trained with rewards like these. DeepSeek-AI, *Nature* 645, 633 (2025).

						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	1
The end-of-answer token: the episode ends						1
						1 The sum of the rewards

Agentic AI, part 2

Agents

Carlos Cotrini

UZH CAS Gamechanger AI

26 September 2026

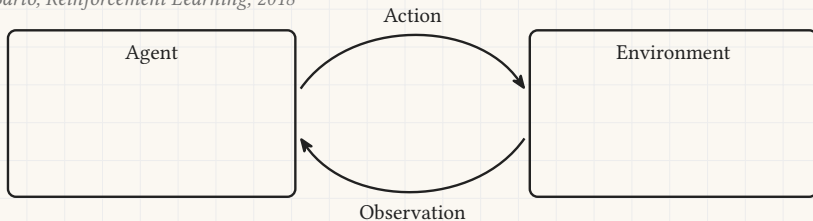
1

Section 1

Agents and the loop

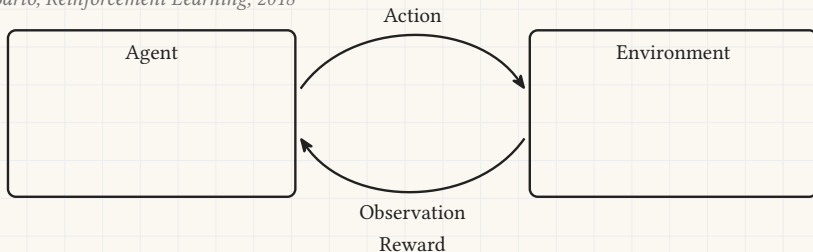
The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



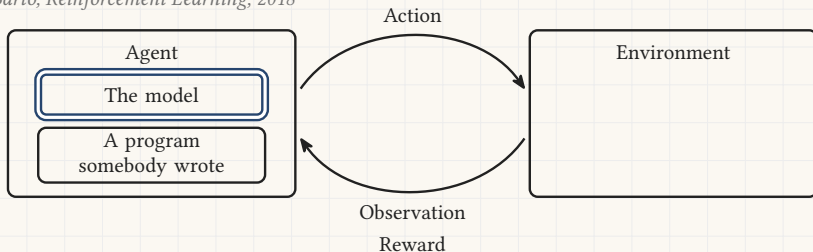
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Sutton and Barto, Reinforcement Learning, 2018



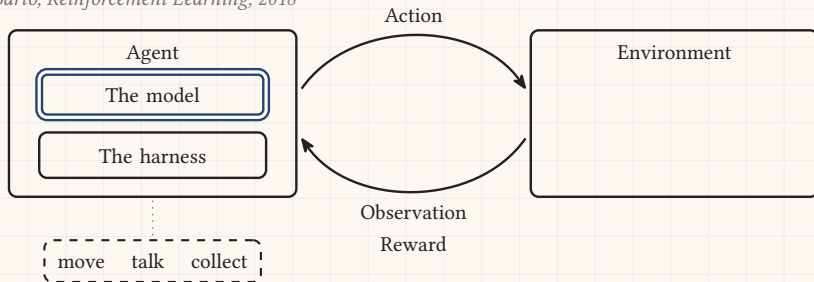
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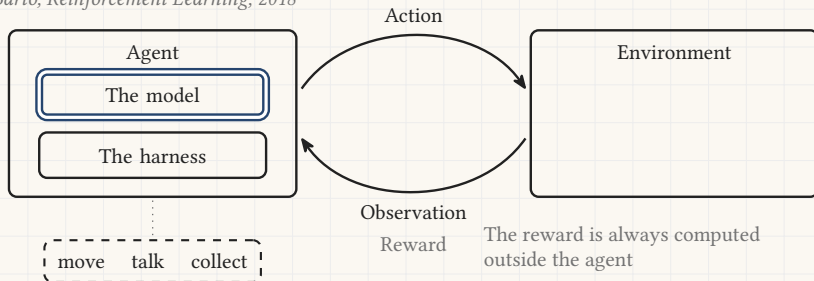
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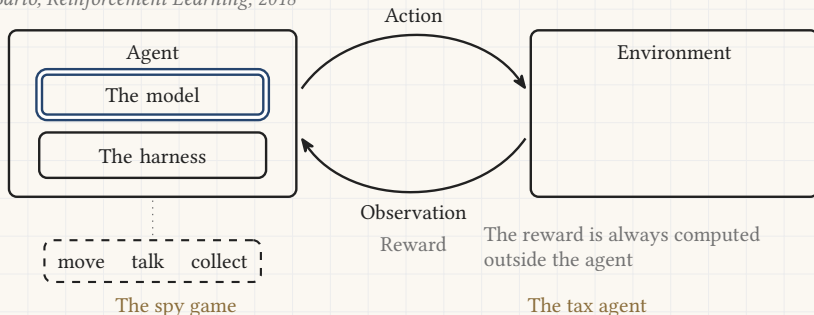
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Sutton and Barto, Reinforcement Learning, 2018



The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



Environment

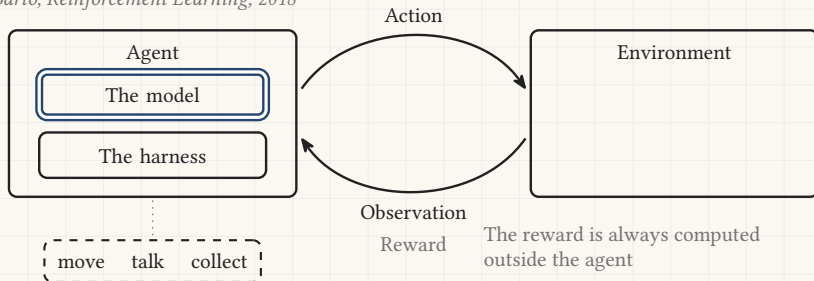
Observations

Actions

Reward

The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The reward is always computed outside the agent

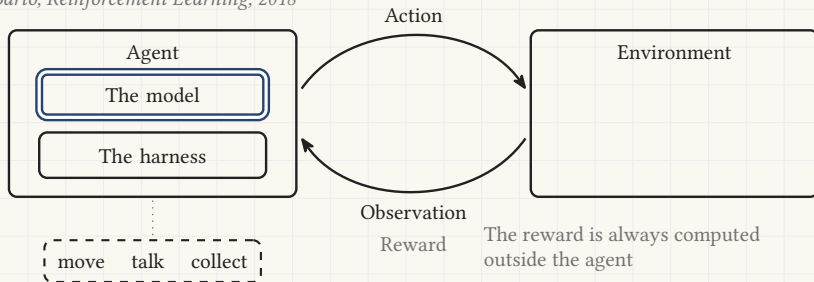
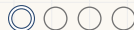
The spy game

The tax agent

Environment	The 3 by 3 base
Observations	What the game prints after each move
Actions	Three tools, and one scan the harness runs
Reward	None is computed

The agent and its environment

Sutton and Barto, Reinforcement Learning, 2018



The spy game

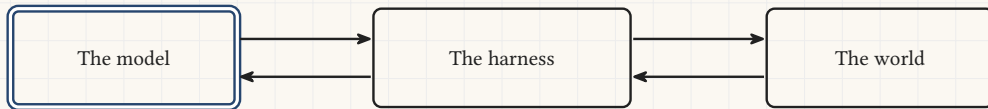
Environment	The 3 by 3 base
Observations	What the game prints after each move
Actions	Three tools, and one scan the harness runs
Reward	None is computed

The tax agent

Environment	Documents, a form, a mail account
Observations	What a tool returns
Actions	Read, extract, fill, send
Reward	None is computed

The harness

The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.

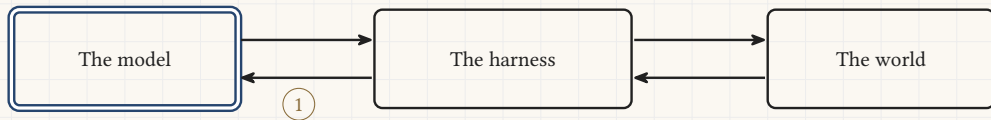


*Anthropic, Claude API documentation,
tool use, 2026*

*Young, Effective harnesses for
long-running agents, Anthropic, 2025*

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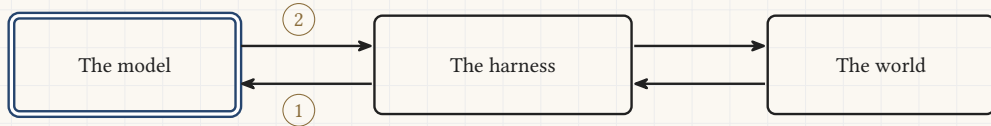
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*Anthropic, Claude API documentation,
tool use, 2026*

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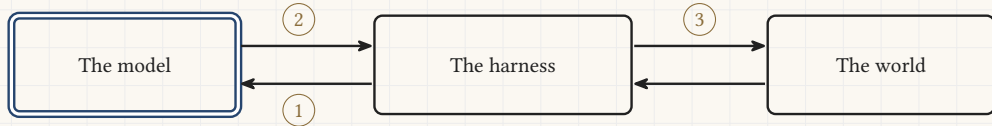
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call

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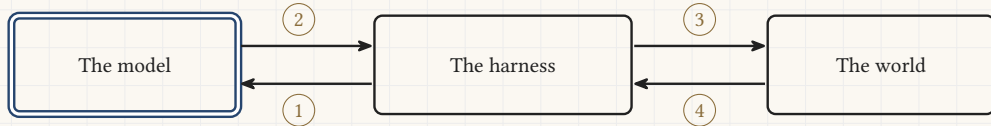
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world

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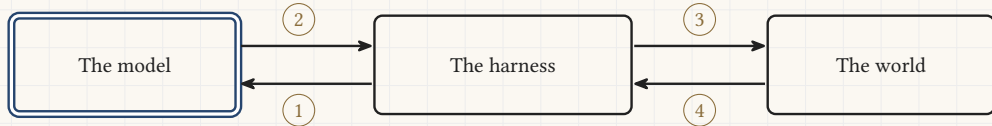
- ① Builds the prompt: the tools and the transcript
- ② Reads the reply and finds the tool call
- ③ Runs that tool on the world
- ④ Appends the result, then calls the model again or stops

*Anthropic, Claude API documentation,
tool use, 2026*

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The harness

The program around the model. The model only writes text; the harness turns that text into actions and feeds back what happened.



In the spy game: `run_agent()`

- 1 Builds the prompt: the tools and the transcript
- 2 Reads the reply and finds the tool call
- 3 Runs that tool on the world
- 4 Appends the result, then calls the model again or stops

*Anthropic, Claude API documentation,
tool use, 2026*

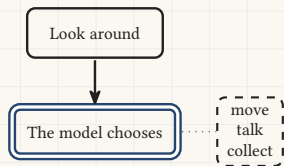
*Young, Effective harnesses for
long-running agents, Anthropic, 2025*

2

Section 2

ReAct, played by hand

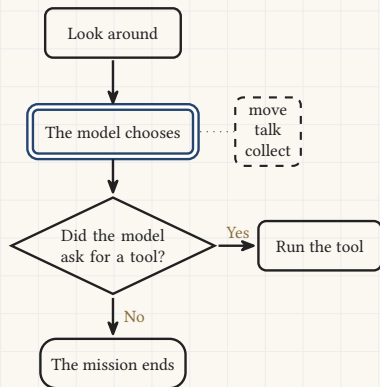
Thought, action, observation



-  A call to the model
-  Something that is not a call to the model
-  The tools the model may choose from

Yao and others, ReAct, ICLR 2023

Thought, action, observation



A call to the model



Something that is not a call to the model



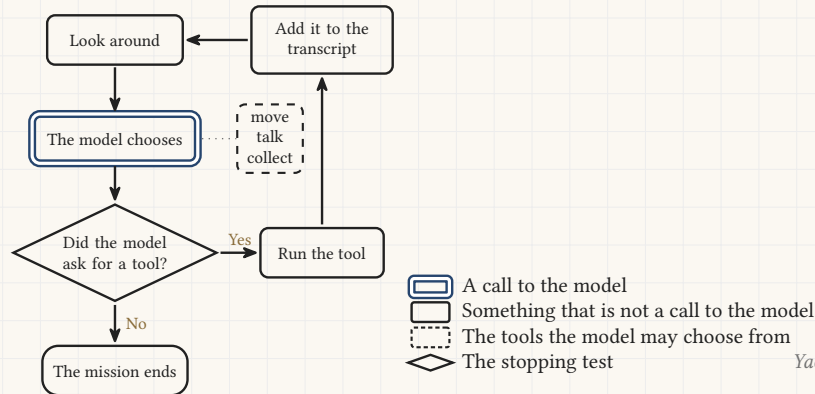
The tools the model may choose from



The stopping test

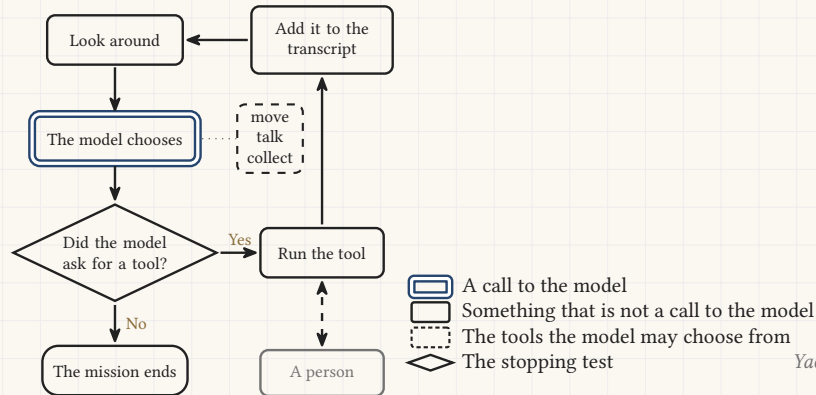
Yao and others, ReAct, ICLR 2023

Thought, action, observation



Yao and others, ReAct, ICLR 2023

Thought, action, observation



Yao and others, ReAct, ICLR 2023

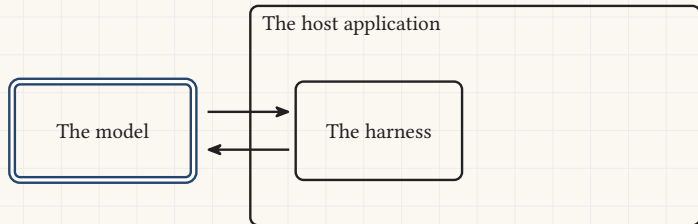
3

Section 3

The Model Context Protocol

What MCP is

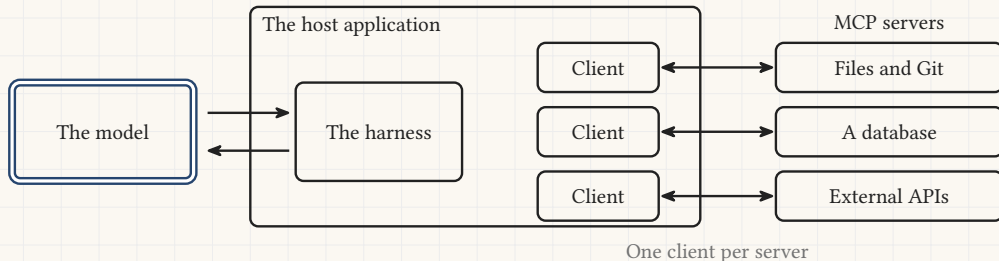
An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Model Context Protocol, Specification, 2026

What MCP is

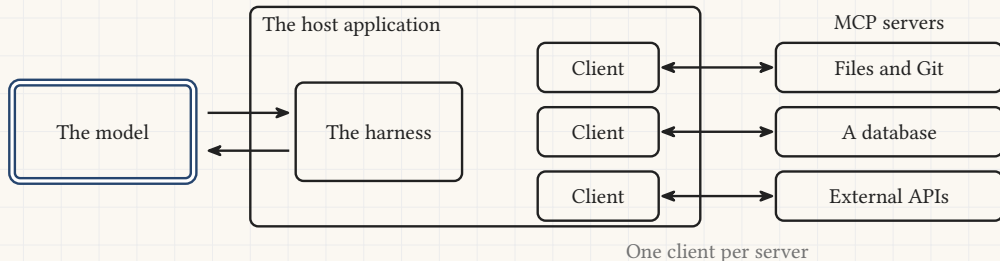
An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Model Context Protocol, Specification, 2026

What MCP is

An open protocol: a standard plug between an application that uses a model and the servers that offer it tools and data.



Tools: functions the model may ask for, run by the server

Resources: context and data, for the user or the model

Prompts: templated messages and workflows for the user

Model Context Protocol, Specification, 2026

4

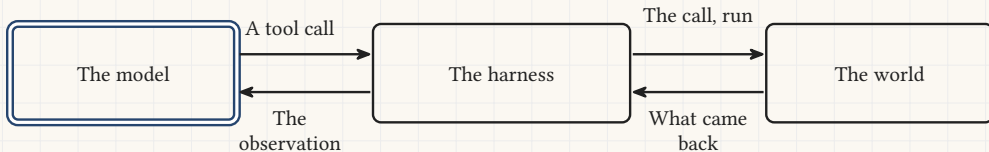
Section 4

The harness

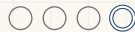
An agent is a system



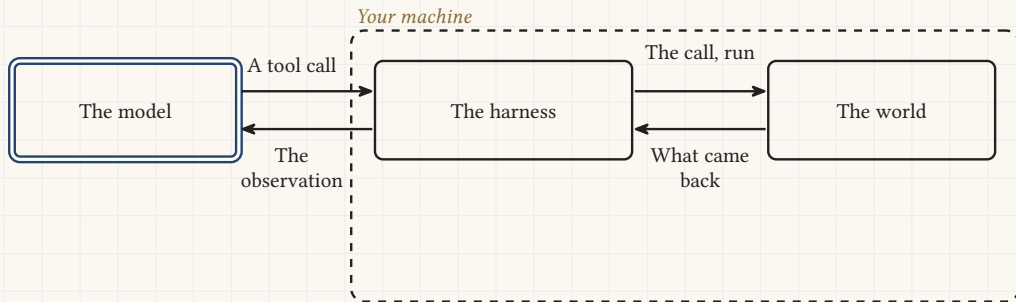
Cherny, Claude Code best practices, Anthropic, 2025
Anthropic, Claude API documentation, tool use, 2026



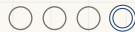
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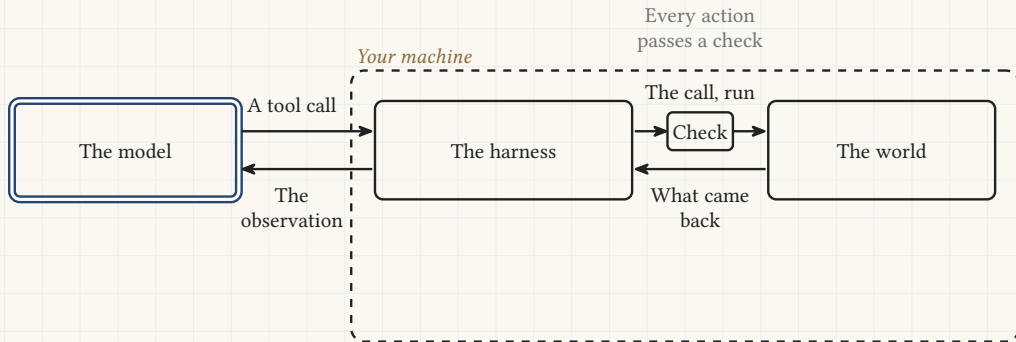
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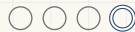
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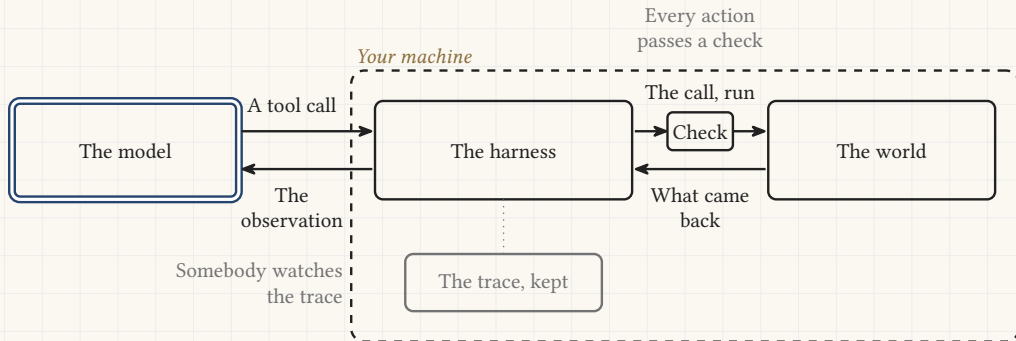
Cherny, Claude Code best practices, Anthropic, 2025
Anthropic, Claude API documentation, tool use, 2026



An agent is a system



Cherny, Claude Code best practices, Anthropic, 2025
Anthropic, Claude API documentation, tool use, 2026



Play the spy game



[https://eth-bmai-hs26.github.io/
BMAI-PAGE/viz/we2/spy-game/](https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we2/spy-game/)

- 1 Open the page in your browser
- 2 Paste your OpenRouter key and press Save
- 3 Play with Manual, or choose Auto and press Run Agent